



The Maths and Neuroscience of Walking

Walking is Not as Easy as You Think

Name: _____ Date: _____

Exercise 1: Sensory Systems

1.1 How many senses are there and name them:

1.2 What is Multisensory Integration and give an example

Exercise 2: Self-motion

2.1 What is Self-motion give an example?

2.2 Why is researching walking important?

2.3 What is the most important sense for walking?

1. We use vision much more than body cues
2. We use body cues much more than vision
3. We use body cues as much as vision

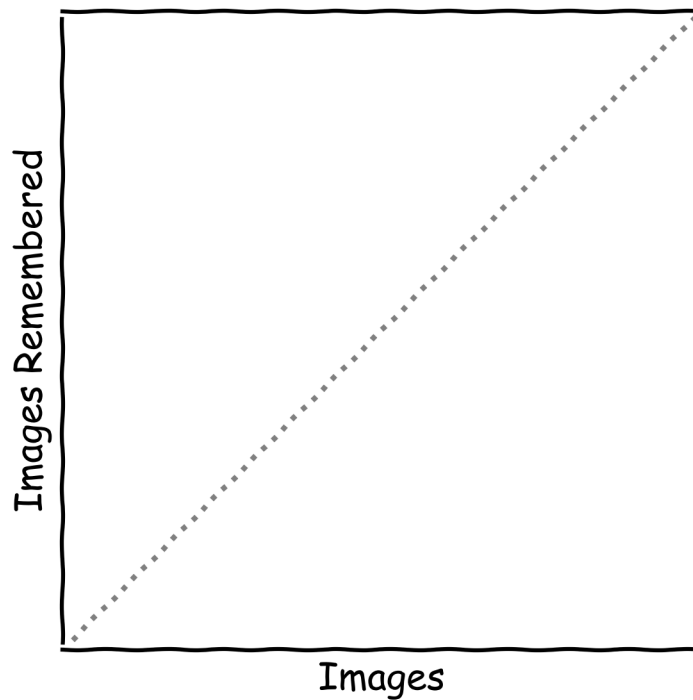
2.4 How to study Walking

How would you isolate vision or motion?



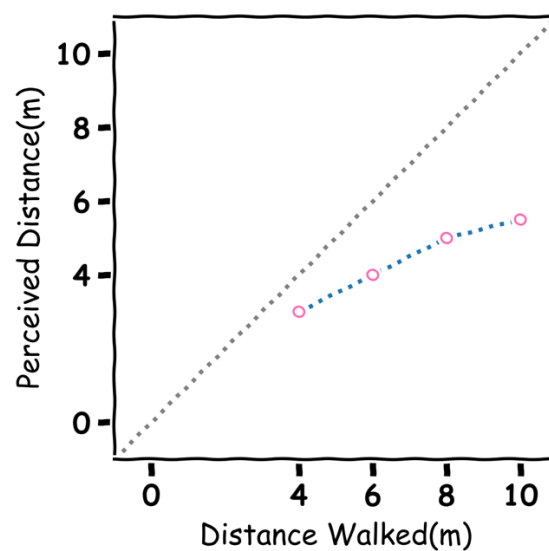
Exercise 3: How Much Do You Remember

Label the graph and add the data:



Exercise 4: Walking Distance

Label the graph and other lines



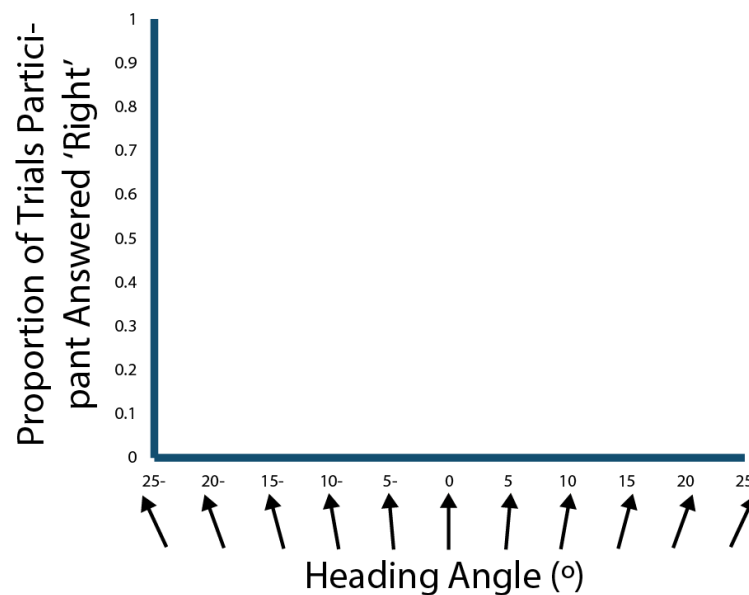


Exercise 5: Left or Right

Exercise 5.1: Fill in the table

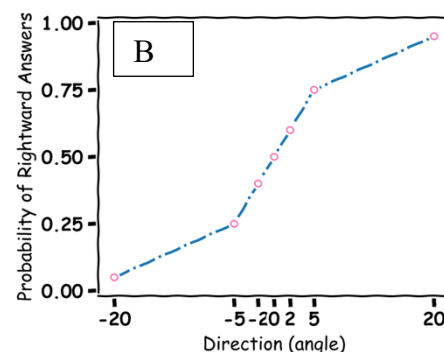
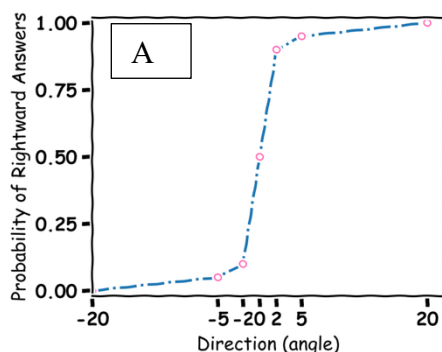
REPITITION	-20	-1	1	20
1				
2				
3				
AVERAGE				

Exercise 5.2: Label the graph and add the data:



Exercise 6: Is A better than B?

Participant A and B did the same heading task from the plots below which participant was better?





Exercise 7: Multisensory Integration

Exercise 7.1 What are the benefits of Multisensory Integration?

Exercise 7.2 Sometimes conflict is good

7.2 Fill in the table

	Vestibular	Visual	Vestibular Weight
Equally			
Vestibular More			
Visual More			

Open Question

What research question would you like to do on Multisensory Integration: